Test Case Document

Student Attendance Management System (SAMS)

Project	Student Attendance Management System (SAMS)
Version	2.0
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Build tested	main branch, commit 33cdb28, plus the new test files
Environment	Python 3.12.3, Django 5.2.18, SQLite, Linux

Revision History

Version	Date	Author	Change
2.0	06-11-2026	Team 1	Updated from the test cases in the Test Plan v1.0 after Sprint 1 and Sprint 2.

Approvals

Role	Name	Signature / Date
Team Lead	Atul Kandiyil	
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1. Introduction

This document is the updated set of test cases for SAMS. The first set was written for the Software Test Plan (16-09-2026) and the acceptance tests in the SRS. Since then we built Sprint 1 and Sprint 2, and a few things came out different from what we planned, so we went through every test case and changed it to match the system that exists now. The test case IDs are the same as before so the traceability matrix in the SRS still lines up.

For each case the table in section 3 shows the steps and expected result, how it is tested (automated tests in the code, and a scenario on the running system), the result from this test round, and what changed from the first version.

2. What changed since version 1.0

- Attendance can now be Present, Absent or Late. The percentage is (Present + Late) / Total.
- The low-attendance limit is 75%, and a student is flagged only when strictly below it. It can be changed with a threshold value in the defaulters page URL.
- Students, faculty, subjects and enrolments are managed in the Django admin portal. We did not build separate pages for them.
- Changing a saved attendance record now writes an audit log entry with the old value, new value and reason. This is a new test case, TC-AUDIT-01.
- Sprint 2 added CSV export, the health check, the report filters and dashboards for each role, so there are new checks for them.
- Performance is tested with our own load script and a 100-thread run on the running server, instead of JMeter.
- The suite went from 20 automated tests to 83.
- Things in the SAD that are not in the SRS (QR and biometric capture, lockout after 5 failed logins, the 48-hour correction window) are not built and not tested. See OBS-02 in the System Validation Report.

3. Test cases

There are 35 test cases. 32 passed. The other 3 are partial, either because of an open defect or because a manual check has not been done yet.

TC ID	Requirement	Title	Steps and expected result	Tested by	Result	Change since v1.0
TC-AUTH-01	SAM-F-001	Valid login	Enter valid username and password on /accounts/login/. A session is created and the user lands on the dashboard for their role.	3 unit, SV-04	Pass	Changed: login now redirects through /role-redirect/ (faculty go to the faculty dashboard).

TC-AUTH-02	SAM-F-002	Invalid login rejected	Wrong password or unknown user. No session, an error is shown, and the message is the same in both cases.	3 unit, SV-03	Pass	Added the same-message check.
TC-AUTH-03	SAM-F-003	Role-based access	Open faculty/admin pages as a student, student pages as faculty, admin pages as admin. 403 outside the role, 200 inside it.	3 unit, SV-07	Pass	Now covers the Sprint 2 pages (reports, audit log, defaulters).
TC-AUTH-04	SAM-F-004	Logout	Log in, log out, open the dashboard. Redirected to login.	1 unit, SV-20	Pass	Changed: logout is a POST request in Django 5.
TC-STU-01	SAM-F-005	Add student	Admin adds a student in the admin portal. Record is saved, duplicate student ID is rejected.	3 unit, SV-19	Pass	Changed: done through the Django admin portal.
TC-STU-02	SAM-F-006	Update student	Admin edits department and year. The change is saved.	1 unit, SV-19	Pass	Changed: done through the Django admin portal.
TC-STU-03	SAM-F-007	View / search students	Admin opens the student list. Faculty and admin filter students by ID or name on the reports page.	2 unit, SV-19, SV-15	Pass	Changed: search is the student filter on the reports page (see OBS-04).
TC-FAC-01	SAM-F-008	Create faculty account	Admin creates a faculty user and profile. That faculty member can log in.	1 unit, SV-19	Pass	Changed: done through the Django admin portal.
TC-SUB-01	SAM-F-009	Manage subjects	Create a subject. It is saved and the subject code must be unique.	1 unit, SV-19	Pass	No change.
TC-FAC-02	SAM-F-010	Assign faculty to subject	Assign faculty to a subject. It shows on that faculty member's dashboard and nobody else's.	2 unit, SV-09	Pass	Added the check that other faculty do not see it.
TC-ENR-01	SAM-F-011	Student enrolment	Enrol students in a subject. Only enrolled students appear in the marking list. A duplicate enrolment is rejected.	2 unit, SV-11	Pass	No change.
TC-ATT-01	SAM-F-012	Mark attendance	Faculty opens a session and marks Present, Absent or Late for every enrolled student. Records are saved and an audit entry is written.	8 unit, SV-10, SV-11	Pass	Changed: Late added as a status; session creation and the mark-all buttons added.
TC-ATT-02	SAM-F-013	Block unauthorised marking	A student, or faculty who do not teach the subject, try to mark attendance (page and direct POST). 403 or 404 and nothing is saved.	3 unit, SV-08	Pass	Added the direct POST and other-faculty cases.
TC-ATT-03	SAM-F-014	Modify attendance	Assigned faculty change a saved status and give a reason. The record is updated and the audit log keeps the old value, new value and reason.	6 unit, SV-13	Pass	Changed: reason is recorded. There is no time limit on modifying (OBS-02).
TC-ATT-04	SAM-F-015	Prevent duplicates	Submit the same session twice, or insert a second record for the same student and session. Only one record exists.	3 unit, SV-10, SV-12	Pass	Added the re-submission case from the UI.
TC-ATT-05	SAM-F-016	Store attendance details	A saved record has the student, session (subject and date), status, who marked it and when.	2 unit, SV-11	Pass	No change.
TC-VIEW-01	SAM-F-017	Student views own attendance	Student opens /attendance/my/ and sees only their own records.	2 unit, SV-05	Pass	Added: a student user with no student profile gets a 404.
TC-VIEW-02	SAM-F-019	Faculty views assigned classes	The faculty dashboard lists only the subjects assigned to that faculty member.	1 unit, SV-09	Pass	No change.
TC-CALC-01	SAM-F-018	Attendance percentage	Percentage = (Present + Late) / Total x 100, 2 decimal places, 0 if there are no classes.	2 unit, SV-05	Pass	Changed: Late counts as attended. Boundary values added.
TC-CALC-02	SAM-F-020	Identify low attendance	Students strictly below 75% (or the ?threshold= value) are listed. Students with no records are not flagged.	4 unit, SV-14	Pass	Changed: threshold is 75% and can be changed in the URL. A bad value falls back to 75.
TC-HIST-01	SAM-F-024	Attendance history	Student sees earlier records, newest first.	1 unit, SV-05	Pass	No change.
TC-REP-01	SAM-F-021	Generate reports	Filter by subject, student and date range. Only matching records show. Faculty see their own subjects, admin sees all.	6 unit, SV-15	Pass	Added date range, student filter and role scoping.
TC-REP-02	SAM-F-022	Summaries and CSV export	The totals on screen match the rows in the CSV export.	2 unit, SV-15	Pass	Changed: CSV export was added in Sprint 2.
TC-NOT-01	SAM-F-023	Low-attendance notification	A student below 75% sees a warning banner. A student above 75% does not.	1 unit, SV-05, SV-06	Pass	Changed: it is an on-screen banner, there is no email or SMS.
TC-PERF-01	SAM-NF-001	Response time (p95 under 3 s)	Key pages answer within 3 seconds and the load script gives p95 of 3 s or less.	1 unit, SV-24	Pass	Changed: our own load script is used instead of JMeter (Deviation 2).
TC-REL-01	SAM-NF-002	Availability / health check	/health/ gives 200 when the database is up and 503 when it is down.	2 unit, SV-01	Pass	New: health endpoint added in Sprint 2.

TC-SEC-01	SAM-SR-001, SAM-NF-003	Login required	Anonymous requests to protected pages are redirected to the login page.	1 unit, SV-02	Pass	Now covers all Sprint 2 URLs.
TC-SEC-02	SAM-SR-002	Roles enforced on the server	Requests from the wrong role (GET and POST) are refused.	2 unit, SV-07, SV-08	Pass	Extended.
TC-SEC-03	SAM-SR-003	Password hashing	Stored passwords are salted PBKDF2 hashes and never plain text. Two users with the same password get different hashes.	1 unit, SV-21	Pass	Changed: the test runner uses MD5 for speed (Deviation 3), so this is checked with PBKDF2 and on the live database.
TC-SEC-04	SAM-SR-004	HTTPS	The proxy SSL header is set and the session and CSRF cookies are marked Secure.	2 unit, SV-22	Partial (DEF-02)	Changed: now checks the cookie flags, which fails (DEF-02). Real TLS on a server was not tested (OBS-05).
TC-AUDIT-01	SAM-NF-006	Audit logging	Creating and modifying records writes an audit entry (user, action, entity, old and new value, reason, time). Only admin can read it.	4 unit, SV-13, SV-18	Pass	New: audit log added in Sprint 2.
TC-SEC-05	SAM-SR-005	Security activity logging	Important actions record the user and a timestamp.	3 unit, SV-18	Pass	Same checks as TC-AUDIT-01.
TC-UX-01	SAM-NF-004	Usability of marking	The mark-all buttons are on the marking page. A faculty member can mark a class without help.	1 unit	Partial (manual trial pending)	Changed: mark-all buttons (Deviation 4). The manual user trial has not been done (OBS-05).
TC-DATA-01	SAM-NF-005	Data integrity	The database refuses duplicates, and a rejected submission leaves no partial data.	2 unit, SV-23	Partial (DEF-01)	Added the atomicity check, which fails (DEF-01).
TC-SCALE-01	SAM-NF-007	100 concurrent users	100 users at the same time with no errors.	SV-24	Pass	Checked with performance_test.py and a 100-thread run on the live server.

"Tested by" shows how many automated tests are linked to the case (listed in the Unit Test Report) and which system scenarios (SV-nn) in the System Validation Report cover it.

4. Traceability

Requirement	Test case(s)	Status
SAM-F-001	TC-AUTH-01	Pass
SAM-F-002	TC-AUTH-02	Pass
SAM-F-003	TC-AUTH-03	Pass
SAM-F-004	TC-AUTH-04	Pass
SAM-F-005	TC-STU-01	Pass
SAM-F-006	TC-STU-02	Pass
SAM-F-007	TC-STU-03	Pass
SAM-F-008	TC-FAC-01	Pass
SAM-F-009	TC-SUB-01	Pass
SAM-F-010	TC-FAC-02	Pass
SAM-F-011	TC-ENR-01	Pass
SAM-F-012	TC-ATT-01	Pass
SAM-F-013	TC-ATT-02	Pass
SAM-F-014	TC-ATT-03	Pass
SAM-F-015	TC-ATT-04	Pass
SAM-F-016	TC-ATT-05	Pass
SAM-F-017	TC-VIEW-01	Pass
SAM-F-018	TC-CALC-01	Pass
SAM-F-019	TC-VIEW-02	Pass
SAM-F-020	TC-CALC-02	Pass

SAM-F-021	TC-REP-01	Pass
SAM-F-022	TC-REP-02	Pass
SAM-F-023	TC-NOT-01	Pass
SAM-F-024	TC-HIST-01	Pass
SAM-NF-001	TC-PERF-01	Pass
SAM-NF-002	TC-REL-01	Pass
SAM-NF-003	TC-SEC-01	Pass
SAM-NF-004	TC-UX-01	Partial
SAM-NF-005	TC-DATA-01	Partial
SAM-NF-006	TC-AUDIT-01	Pass
SAM-NF-007	TC-SCALE-01	Pass
SAM-SR-001	TC-SEC-01	Pass
SAM-SR-002	TC-SEC-02	Pass
SAM-SR-003	TC-SEC-03	Pass
SAM-SR-004	TC-SEC-04	Partial
SAM-SR-005	TC-SEC-05	Pass

All 36 of the 36 requirements in the SRS have at least one test case. 33 are passing and 3 are partial: SAM-NF-005 (DEF-01), SAM-SR-004 (DEF-02) and SAM-NF-004 (manual usability trial not done).

5. Test data

Demo data comes from `python manage.py seed_demo`: admin, prof_smith, alice, bob and charlie, with subjects CS201 and CS301. Bob is a defaulter at 50% and charlie is at 100%. The unit tests make their own data in each test and use a temporary database, so they do not depend on each other.

6. Running the tests

```
python manage.py test -v 2              (automated tests)
python scripts/performance_test.py      (load test)
python scripts/system_validation.py     (runs against a live server)
```